Analysis of CP Asymmetry in $B^{\pm} \to \pi^{\pm}\pi^{+}\pi^{-}$ Decays

Third Year Lab Report

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This study investigates global and localised CP violation in the charmless three-body decay $B^{\pm} \to \pi^{\pm}\pi^{\pm}\pi^{\mp}$ using data from the LHCb experiment. Following event selection optimisation and charm resonance vetoes, a simultaneous extended maximum likelihood fit to the $B^{\pm}$ invariant mass distributions yields a global raw asymmetry of $A_{\text{raw}} = 0.07 \pm 0.03$. Local CP asymmetry is subsequently analysed across the Dalitz phase space using two independent background subtraction methods, the sPlot technique and a sideband subtraction modelled with 2D Chebyshev polynomials. Both methods consistently identify highly significant ($> 3\sigma$) localised CP asymmetries, most notably in the $\rho^0(770)$ resonance region ($m^2_{\text{low}} < 1$ GeV$^2/c^4$) and the $\pi\pi \leftrightarrow KK$ rescattering band ($1 < m^2_{\text{low}} < 2.25$ GeV$^2/c^4$). The agreement between these independent methods shows that sideband subtraction is an effective alternative to sPlot for cross-validating local asymmetries.

1. INTRODUCTION

CP violation occurs when the laws of physics are not invariant under charge conjugation (C) and parity (P) operations. It is one of the three Sakharov conditions required to explain the matter-antimatter asymmetry in the universe [1]. While the Standard Model accommodates the CP violation via the Cabibbo-Kobayashi-Maskawa (CKM) matrix [2], this effect is insufficient to explain the observed cosmological matter-antimatter asymmetry, which motivates the search for new physics [3].

Large direct CP violation has been observed in charmless 3-body decays of B mesons, such as into three pions, but detailed hadronic mechanisms are still not fully understood [4]. Precision analysis of both global and local asymmetry of these decays can help reveal the underlying mechanics and potential new physics. In this experiment, $B^{\pm} \to \pi^{\pm}\pi^{+}\pi^{-}$ from LHCb data was analysed to study global and local CP violation. $B^{\pm}$ mesons decay into these final states by weak interaction via many intermediate resonance states before the final three pions states.

The raw matter-antimatter asymmetry is defined as

$$A_{\text{raw}} = \frac{N^- - N^+}{N^- + N^+} \tag{1}$$

where $N^{\pm}$ is the observed $B^{\pm}$ decay events. This measured value includes the true CP asymmetry, but it can also be faked by the production asymmetry of $B^{\pm}$ mesons and the detection asymmetry of product particles [5].

The global asymmetry is determined by reconstructing the invariant mass $s = (\sum \tilde{p}_i)^2$, where $\tilde{p}_i$ denotes the 4-momentum of particles detected in the final-state. By calculating the invariant mass under the assumption that all tracks are pions, true B meson decays can be distinguished from background events.

Local asymmetry is analysed using a Dalitz plot, which fully describes the kinematics of 3-body decays using only two variables, the squared invariant masses of two particles pairs (m^2_{12}, m^2_{23}). This 2D kinematic phase space reveals localised intermediate structures such as resonances and interference patterns [6]. To resolve the inherent assignment ambiguity of the two identical-sign pions in the final state, the Dalitz coordinates were conventionally defined as the smaller (m^2_{low}) and greater (m^2_{high}) squared invariant masses of the two possible opposite-sign pion pairs."

2. METHODOLOGY

2.1. The LHCb Detector & Data

The LHCb detector is a single-arm forward spectrometer designed for the study of particles containing b or c quarks [7]. The tracking system consists of a Vertex Locator (VELO) and four planar tracking stations made of silicon microstrip. This enables reconstructing the primary pp vertex and the displaced secondary vertex of b-hadrons, which are crucial to distinguish the targeted decays from the background. The final state hadrons are identified using two Ring-Imaging Cherenkov (RICH) detectors, which provide particle identification (PID) of pions and kaons.

The momenta and charges of the decay products are determined by the tracking system combined with a dipole magnet with a bending power of 4 Tm [8]. This system provides a relative momentum resolution ranging from 0.5 % at low momentum below 20 GeV/c to 0.8 % at $\tilde{1}$00 GeV/c. Because charged particles are bent in a specific direction, it can induce a fake CP asymmetry due to geometric detection asymmetry. To control this systematic effect, the polarity of the magnetic field was regularly reversed during data acquisition, allowing the geometric detection asymmetry to be cancelled out.

2.2. Data Selections

To ensure a precise measurement of the CP asymmetry, it is necessary to filter the pre-selected data to maximise the statistical significance of the targeted signal. The optimal selection criteria were determined by maximising the signal-to-noise ratio (SNR) defined as $N_S/\sqrt{N_S + N_B}$, where N_S and N_B are the estimated number of signal and combinatorial background events, respectively.

Prior to the optimisation procedure, basic data quality cuts were first applied, and candidates containing muon tracks were removed. The selection was then optimised using a grid search over three variables chosen to exploit the decay topology of $B^{\pm}$. The first is the impact parameter (IP) χ^2, which measures the displacement of a track from the primary vertex [8]. The second is the secondary vertex fit χ^2, which mea-

sures how well the three candidate pion tracks geometrically converge. Finally, the particle identification variable $\mathrm{PID}_{\pi K} = \mathcal{P}(\pi) - \mathcal{P}(K)$, defined as the probability difference between the pion and kaon hypotheses, was used to isolate pion signals from kaon backgrounds.

For each parameter combination in the grid search, N_S and N_B were estimated by fitting the reconstructed invariant mass distribution. The signal was modelled with a Gaussian distribution and the background with an exponential function over the mass range [5150, 5600] MeV/c^2.

The optimal requirements were found to be $\mathrm{PID}_{\pi K} > -0.389$ and IP $\chi^2 > 27.56$. No additional constraint on the vertex χ^2 was needed beyond the pre-selection requirement of < 12. Although the fit model was not ideal, the optimal thresholds were insensitive to small modelling inaccuracies.

2.3. Charm Resonance Vetoes

Because this analysis targets charmless $B^\pm$ decays, any true $B^\pm$ events that proceed via an intermediate charm resonance must be removed. The primary sources of this contamination are $B^\pm$ decays involving intermediate D mesons, which can yield true three-pion final states or states where a kaon is misidentified as a pion. Another source is J/ψ resonances where muons are misidentified as pions. Although the probability of such misidentifications is low, these channels have much larger branching fractions and can therefore significantly contaminate the signal.

To eliminate these charmed B decay modes, the invariant masses of any possible two-body tracks were recalculated under the respective particle mass hypotheses. By fitting these distributions with a Gaussian signal and a linear background, the centres and widths (σ) of the resonances were identified. Events falling within a $\pm 3\sigma$ window of the Gaussian peaks were vetoed subsequently.

2.4. Global Asymmetry Method

To determine A_{raw}, an extended maximum likelihood fit was used to obtain the signal yields of the $B^\pm$ decays. The likelihood $\mathcal{L}$ of observing a data set of N total events with invariant masses $\{m_i\}$, given a set of parameters $\boldsymbol{\theta}$, is defined as

$$\mathcal{L} = e^{-\sum_j N_j} \prod_{i=1}^{N} \left(\sum_k N_k \mathcal{P}_k(m_i; \boldsymbol{\theta}_k) \right), \qquad (2)$$

where N_k is the expected yield of component k, and $\mathcal{P}_k$ is its probability density function (PDF) [9]. The yields and shape parameters were then determined by minimising $-2\ln\mathcal{L}$. To avoid additional biases on the asymmetry, the fit was performed simultaneously on the B^+ and B^- datasets. While $\boldsymbol{\theta}$ were shared and optimised in both datasets, $N^\pm$ were allowed to be optimised independently.

2.5. Local CP Asymmetry Methods

To ensure a rigorous statistical analysis of the local CP asymmetry across the Dalitz plot, two distinct background subtraction methods were performed and compared.

2.5.1. sPlot Technique

The sPlot technique is a standard statistical method used to isolate the signal from the background in the Dalitz plot [10]. The weight of each data is given by

$$w_i^{(S)} = \sum_k V_{Sk} \frac{\mathcal{P}_k(m_i; \boldsymbol{\theta}_k)}{\sum_j N_j \mathcal{P}_j(m_i; \boldsymbol{\theta}_j)}, \qquad (3)$$

where V_{Sk} is the covariance matrix element between the signal yield S and the component k. This is calculated using the probability density functions and yield covariance matrix derived from the reconstructed invariant mass fit [9]. By applying a signal weight (sWeight) to the dataset, the background contributions are statistically subtracted to leave a pure signal Dalitz plot. A critical assumption of this technique is that the invariant mass and the Dalitz plot coordinates are uncorrelated.

2.5.2. Sideband Subtraction Technique

As a complementary approach, the combinatorial background was modelled using data from the invariant mass sideband regions and subsequently scaled to match the expected background yield in the signal region before subtraction. The reconstructed invariant mass fit performed for the global CP asymmetry measurement was used for this procedure. The background Dalitz plot was modelled using a 2D basis of Chebyshev polynomials of the first kind as

$$\hat{B}(x', y'; c_{ij}) = \sum_{i=0}^{d_{\mathrm{low}}} \sum_{j=0}^{d_{\mathrm{high}}} c_{ij} T_i(x') T_j(y') \qquad (4)$$

where T_i and T_j are the Chebyshev polynomials of degree i and j, c_{ij} are the corresponding model coefficients, and d_{low} and d_{high} denote the maximum polynomial degrees along the $(m^2_{\mathrm{low}}, m^2_{\mathrm{high}})$ axes, respectively. To minimise coefficient correlations, the variables were linearly mapped to the $[-1, 1]$ domain, and the model coefficients c_{ij} were determined via Ordinary Least Squares (OLS) minimisation. A detailed discussion of the error estimation associated with this polynomial modelling can be found in Ref. [11].

The optimal maximum polynomial degrees were determined by minimising the reduced χ^2 of the 1D projections onto each Dalitz axis. To prevent overfitting, an early stopping criterion was adopted. This optimisation yielded final maximum degrees of $(d_{\mathrm{low}}, d_{\mathrm{high}}) = (7, 5)$.

3. RESULTS

3.1. Global CP Asymmetry

Figure 1 presents the reconstructed invariant mass distributions for B^- and B^+ candidates alongside the simultaneous global CP asymmetry fit. The data were modelled using three primary components, the $B^\pm \to \pi^\pm\pi^\pm\pi^\mp$ signal, combinatorial background, and a partially reconstructed background which predominantly arises from four-body decays.

To capture the asymmetric tail, the signal was modelled using a Crystal Ball function. The unstructured combinatorial background was modelled with a simple exponential decay function. Theoretically, the partially reconstructed background should be modelled by an ARGUS function convolved with a Gaussian to

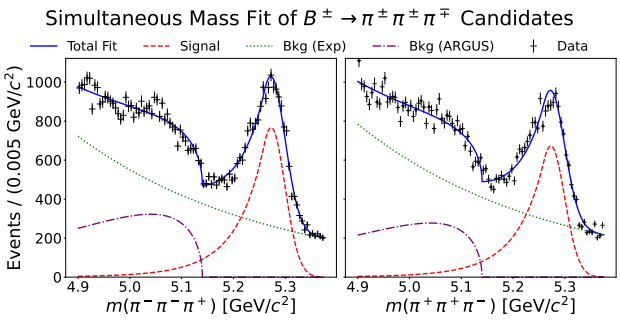

FIG. 1. Simultaneous extended maximum likelihood fit to the reconstructed invariant mass distributions of B^- (left) and B^+ (right) candidates. The data are shown as black points with statistical error bars. The total fit is represented by the solid blue line. The individual fit components are overlaid: the Crystal Ball signal (dashed red), the exponential combinatorial background (dotted green), and the ARGUS partially reconstructed background (dash-dotted purple).

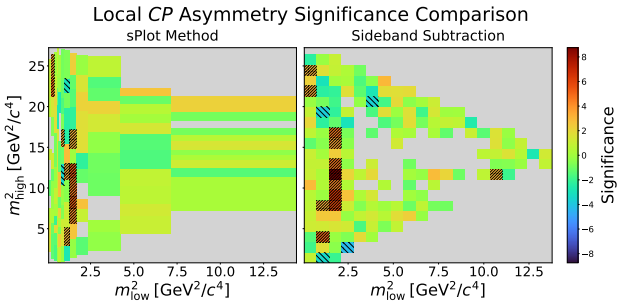

FIG. 2. Local CP asymmetry significance across the $B^{\pm} \to \pi^{\pm}\pi^{\pm}\pi^{\mp}$ Dalitz plot using the adaptive binning sPlot method (left) and the uniform grid sideband subtraction method (right). The axes represent the lesser ($m^2_{\rm low}$) and greater ($m^2_{\rm high}$) squared invariant masses of the opposite-sign pion pairs. Bins with a significance exceeding 3σ are highlighted with overlaid hatching.

account for limited momentum resolution. However, as a simplifying assumption in this analysis, this component was modelled solely using an ARGUS function. This function remains sufficient to capture the physical, kinematic threshold of the four-body decays [12].

From the extracted signal yields, the global raw asymmetry was measured to be $A_{\rm raw} = 0.07 \pm 0.03$ with a statistical significance of 2.3σ. While the statistical uncertainty is relatively large, it is consistent with recent precision measurements of $A_{\rm CP} = 0.080 \pm 0.004$ [4].

3.2. *Local CP Asymmetry*

Figure 2 shows the localised CP asymmetry significance across the Dalitz phase space. The left plot uses the sPlot background subtraction method, while the right plot uses the uniform grid sideband subtraction method. In both plots, regions with a significance larger than 3σ are hatched. The high-significance regions are spatially connected and consistent across both methods.

At low invariant masses ($m^2_{\rm low} < 1$ GeV$^2/c^4$), a prominent region of positive asymmetry appears. This corresponds to the $\rho^0(770)$ resonance [13]. Another concentrated vertical band of high CP asymmetry is visible in the $1 < m^2_{\rm low} < 2.25$ GeV$^2/c^4$ region, which marks the $\pi\pi \leftrightarrow KK$ rescattering region [4]. Neither method reproduces the $\chi_{c0}(1P)$ charmonium peak, which is expected in the high-mass region between $m^2_{\rm high} = 10$ and 15 GeV$^2/c^4$.

Overall, both methods successfully reproduce the dominant localised asymmetries. This demonstrates that while sPlot is the standard statistical tool, sideband subtraction remains a robust alternative for cross-validation in charmless three-body decays.

4. CONCLUSION

In this experiment, CP violation in $B^{\pm} \to \pi^{\pm}\pi^{\pm}\pi^{\mp}$ decays was investigated using LHCb data. To isolate the signal, data selection was optimised for the highest signal-to-noise ratio, and backgrounds from intermediate charm resonances were vetoed. A simultaneous extended maximum likelihood fit to the reconstructed invariant mass distribution measured a global raw asymmetry of $A_{\rm raw} = 0.07 \pm 0.03$, corresponding to a statistical significance of 2.3σ.

Localised CP asymmetries were also evaluated across the Dalitz phase space by comparing two distinct background subtraction techniques, sPlot method and a sideband subtraction method. Both approaches successfully and consistently identified regions of high CP asymmetry ($> 3\sigma$), specifically corresponding to the $\rho^0(770)$ resonance and the $\pi\pi \leftrightarrow KK$ rescattering region. The spatial consistency of these high-significance regions across both models demonstrates that sideband subtraction is a effective alternative to sPlot for cross-validating local asymmetries in charmless three-body decays.

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